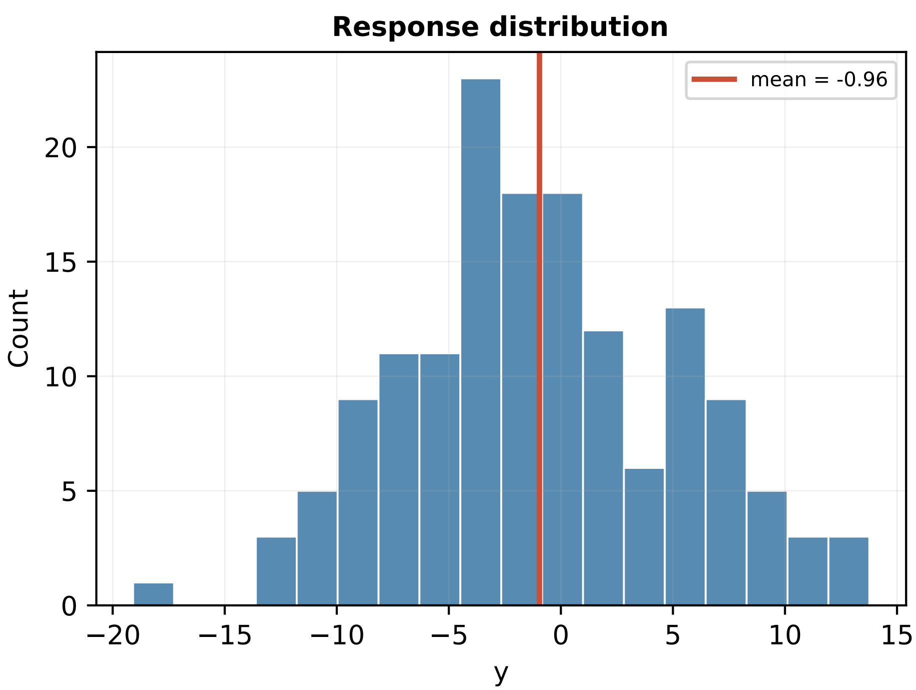
Data

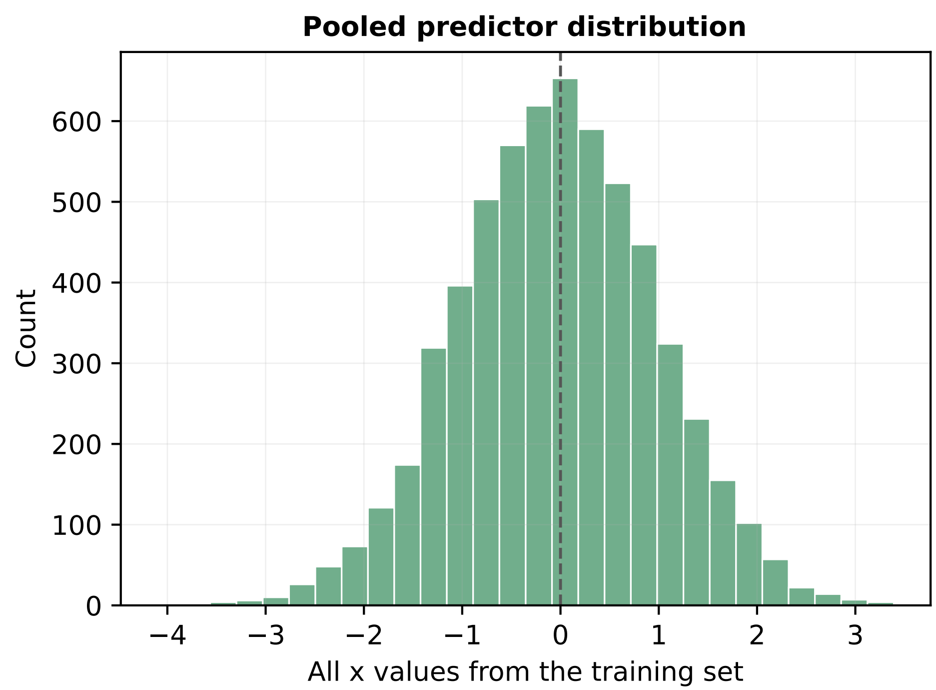
Sample size and summary statistics

Quantity	Value	Notes
train rows	150	observed y
test rows	100	hidden y
predictors	40	x1 through x40
y mean / SD	-0.955 / 6.159	range -19.09 to 13.76
train x mean	-0.035	feature means -0.17 to 0.07
train x SD	0.999	feature SDs 0.86 to 1.11
test x mean	-0.076	similar scale to train

Largest marginal correlations with y

Variable	corr(x,y)	abs corr
x12	-0.456	0.456
x31	0.445	0.445
x1	0.444	0.444
x34	0.392	0.392
x8	0.387	0.387
x7	0.335	0.335
x4	0.331	0.331
x20	-0.286	0.286





Model, Method and Results

Model formula

Ridge minimizes:
(1/n) sum_i (y_i - beta_0 - x_i'beta)^2 + lambda sum_j beta_j^2

Lasso uses the same squared-error loss but replaces the penalty with alpha sum_j |beta_j|.
The intercept is fitted separately and is not penalized.

Method

Ridge: closed-form matrix solve plus explicit gradient descent.
CV: manual 5-fold validation over the provided 80-lambda grid.
Ridge choice: select the lambda with the smallest average MSE across all validation folds.
Scaling: every ridge fold standardizes x using that fold's training rows only.
Lasso: sklearn.linear_model.LassoCV with standardized predictors.

Numerical results

Result	Value
ridge lambda	0.232857
min ridge CV MSE	13.6666
GD vs closed-form RMSE	2.25e-11
lasso alpha	0.211718
lasso CV MSE	12.6459
lasso nonzero betas	17 of 40
ridge test pred mean	-1.059
lasso test pred mean	-0.924

Selected variables

Lasso selected 17 variables:
x1, x3, x4, x7, x8, x12, x15, x18, x19, x20, x24, x26, x29, x31, x34, x35, x38

Ridge keeps all 40 coefficients but shrinks them toward zero.
Lasso sets some coefficients exactly to zero, giving a sparse model.

Model assumptions

Rows are treated as independent training examples.
Train and hidden test rows are assumed to come from the same data-generating process.
The conditional mean is approximated by a linear combination of standardized predictors.
Correlated predictors may make lasso variable selection unstable.

Libraries

Ridge implementation: numpy matrix operations and loops only.
Lasso implementation: sklearn.linear_model.LassoCV.
Data and output: pandas, json.
Slides and figures: matplotlib.
Bootstrap stability: B = 200 lasso refits.

Visualization